

GENERAL CHEMISTRY – CHE 1201

Chemistry of Elements

Introduction and Alkali Metals



What is Chemistry, and What Can Chemistry Do for You?

You may wonder why

- Why the boiling water bubbles and produces steam while the teakettle retains its original shape?
- How do the tea leaves change the color of the water while the teabag remains as full and plump as ever?
- Why does sugar make your tea sweet, and why is the tea itself bitter?
- How gasoline burns and drives vehicles down the road?
- How does it pollute the air we breathe?
- Why some books that are hundreds of years old are still in good shape while other books that are only 50 years old have pages that are brown, brittle, and crumbling.
- Can the books with damaged pages be saved?

An introduction to Chemistry – Mark Bishop

Chemists can answer all these questions
and
Others like them



freegreatpicture.com



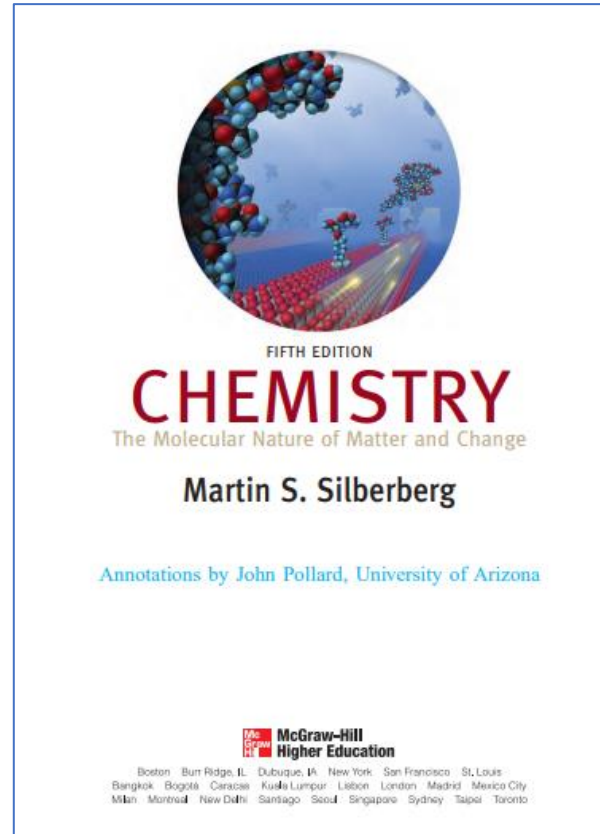
Dreamstime.com



Mrgranato.wordpress.com



Referencing Text



Periodic Patterns in the Main-Group Elements

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Period ↓	1 1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba	* 71 Lu	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra	* 103 Lr	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og
			* 57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb		
			* 89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No		

14.1 Hydrogen, the Simplest Atom

Highlights of Hydrogen Chemistry

14.2 Trends Across the Periodic Table: The Period 2 Elements

14.3 Group 1A(1): The Alkali Metals

14.4 Group 2A(2): The Alkaline Earth Metals

14.5 Group 3A(13): The Boron Family

Highlights of Boron Chemistry

14.6 Group 4A(14): The Carbon Family

Highlights of Carbon Chemistry

Highlights of Silicon Chemistry

14.7 Group 5A(15): The Nitrogen Family

Highlights of Nitrogen Chemistry

Highlights of Phosphorus Chemistry

14.8 Group 6A(16): The Oxygen Family

Highlights of Oxygen Chemistry

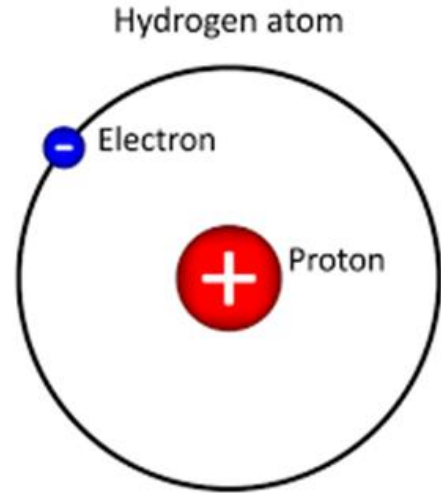
Highlights of Sulfur Chemistry

14.9 Group 7A(17): The Halogens

Highlights of Halogen Chemistry

14.10 Group 8A(18): The Noble Gases

14.1 HYDROGEN, THE SIMPLEST ATOM



- Nucleus with a single positive charge
- Surrounded by a single electron

Importance of hydrogen

- Despite this simple structure, or perhaps because of it, hydrogen may be the most important element of all.
- In the Sun, hydrogen (H) nuclei combine to form helium (He) nuclei in a process that provides nearly all Earth's energy.
- About 90% of all the atoms in the universe are H atoms, making it the most abundant element by far.
- On Earth, only tiny amounts of the free element occur naturally (as H_2), but hydrogen is abundant in combination with oxygen in water.

Properties of H_2 gas

- Colorless
- Odorless
- Very low melting ($-259\text{ }^{\circ}\text{C}$)
- Boiling points ($-253\text{ }^{\circ}\text{C}$).

Where Does Hydrogen Fit in the Periodic Table?

Hydrogen has no perfectly suitable position in the periodic table

- Some think it fits in Group 1A(1) because of its single valence electron and common 1 oxidation state, **However**,
 - Unlike the alkali metals, hydrogen shares its electron with nonmetals rather than losing it to them.
 - Like a nonmetal, it has a relatively high ionization energy (IE 1311 kJ/mol) much higher than that of lithium (IE 520 kJ/mol), whose IE is the highest in Group 1A
 - Much higher electronegativity (EN of H 2.1; EN of Li 1.0).

	1A	2A	3A	4A	5A	6A	7A	8A
	(1)	(2)	(13)	(14)	(15)	(16)	(17)	(18)
1	H			H			H	
2								
3								
4								
5								
6								
7								

Where Does Hydrogen Fit in the Periodic Table?

Hydrogen has no perfectly suitable position in the periodic table

- Others think hydrogen fits in Group 7A(I7) because like the halogens it forms diatomic molecules and fills its outer (valence) level either by electron sharing or by gaining one electron from a metal to form a monatomic anion, the hydride ion (H^- ; O.N. = -1).
- However, hydrogen has a lower electronegativity ($\text{EN} = 2.1$) than any of the halogens (ENs range from 2.2 to 4.0)
- Lacks their three valence electron pairs.
- H^- ion is rare and reactive, whereas halide ions are common and stable.

	1A	2A	3A	4A	5A	6A	7A	8A
	(1)	(2)	(13)	(14)	(15)	(16)	(17)	(18)
1	H			H			H	
2								
3								
4								
5								
6								
7								

Where Does Hydrogen Fit in the Periodic Table?

Hydrogen has no perfectly suitable position in the periodic table

- Still others think hydrogen fits best in Group 4A(14) because of a half-filled valence level and similarities in ionization energy, electron affinity, electronegativity, and bond energies.

Hydrogen's unique behavior is attributable to its tiny size.

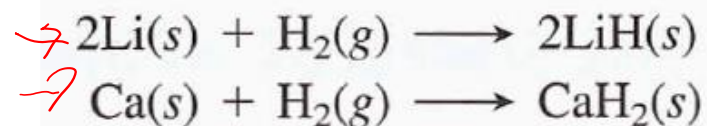
- Hydrogen has a high IE because its electron is very close to the nucleus, with no inner electrons to shield it from the positive charge.
- It has a low EN (for a nonmetal) because it has only one proton to attract bonding electrons.
- In this chapter, hydrogen will appear in either Group 1A(1) or 7A(17) depending on the property being considered.

	1A	2A	3A	4A	5A	6A	7A	8A
	(1)	(2)	(13)	(14)	(15)	(16)	(17)	(18)
1	H			H			H	
2								
3								
4								
5								
6								
7								

Highlights of Hydrogen Chemistry

Ionic (Saltlike) Hydrides

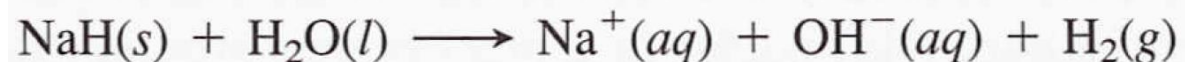
- With very reactive metals, such as those in Group 1A(1) (Li, Na, K) and the larger members of Group 2A(2) (Ca, Sr, and Ba), hydrogen forms *saltlike hydrides*-white, crystalline solids composed of the metal cation and the hydride ion:



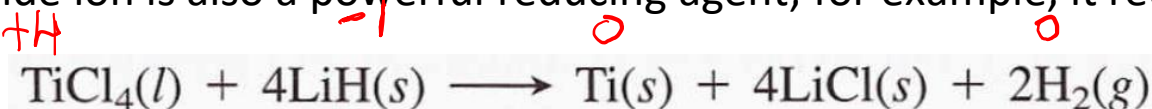
KI

	1A	2A	3A	4A	5A	6A	7A	8A
	(1)	(2)	(13)	(14)	(15)	(16)	(17)	(18)
1	H			H			H	
2								
3								
4								
5								
6								
7								

- In water, H^- is a strong base that pulls H^+ from surrounding H_2O molecules to form H_2 and OH^- :



- The hydride ion is also a powerful reducing agent; for example, it reduces Ti(IV) to the free metal:



Highlights of Hydrogen Chemistry

Covalent (Molecular) Hydrides

- Hydrogen reacts with nonmetals to form many covalent hydrides, such as CH_4 , NH_3 , H_2O and HF .
- In most covalent hydrides, hydrogen has an oxidation number of +1 because the other nonmetal has a higher electronegativity than hydrogen.
- Conditions for preparing the covalent hydrides depend on the reactivity of the other nonmetal.
 - For example, with stable, triple-bonded N_2 , hydrogen reacts at high temperatures ($\sim 400^\circ\text{C}$) and pressures ($\sim 250\text{ atm}$), and the reaction needs a catalyst to proceed at any practical speed:



- hydrogen combines rapidly with reactive, single-bonded F_2 , even at extremely low temperatures (-196°C):

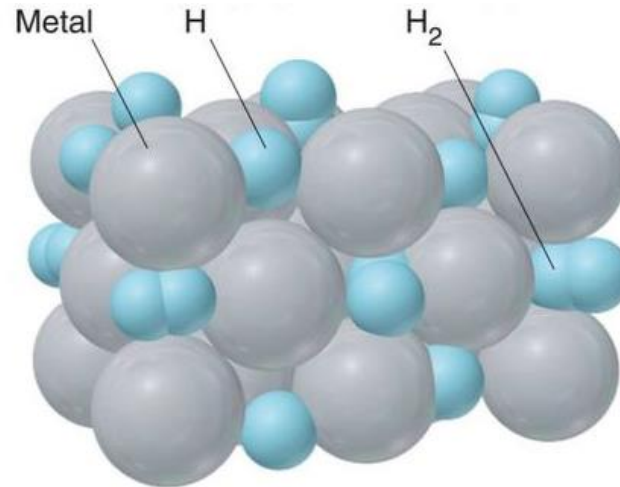


	1A	2A	3A	4A	5A	6A	7A	8A
	(1)	(2)	(13)	(14)	(15)	(16)	(17)	(18)
1	H			H			H	
2								
3								
4								
5								
6								
7								

Highlights of Hydrogen Chemistry

Metallic (Interstitial) Hydrides

- Many transition elements form metallic (interstitial) hydrides, in which H_2 molecules (and H atoms) occupy the holes in the metal's crystal structure. Thus, such hydrides are not compounds but gas-solid solutions.



Interstitial Hydrides

- Also, unlike ionic and covalent hydrides, interstitial hydrides, such as $\text{TiH}_{1.7}$ typically lack a single, stoichiometric formula because the metal can incorporate variable amounts of hydrogen, depending on the pressure and temperature of the gas.

14.2 TRENDS ACROSS THE PERIODIC TABLE: THE PERIOD 2 ELEMENTS

- Electrons fill the one ns and the three np orbitals according to Pauli's exclusion principle and Hund's rule.
- As a result of increasing nuclear charge and the addition of electrons to orbitals of the same energy level (same n value), atomic size generally decreases, whereas first ionization energy and electronegativity generally increase (see bar graphs on p. 557).
- Metallic character decreases with increasing nuclear charge as period members change from metals to metalloids to nonmetals.
- Reactivity is highest at the left and right ends of the period, except for the inert noble gas.
- Bonding between atoms of an element changes from metallic, to covalent in networks, to covalent in individual molecules, to none (noble gases exist as separate atoms). As expected, physical properties, such as melting point, change abruptly at the network/molecule boundary, which occurs in Period 2 between carbon (solid) and nitrogen (gas).
- Bonding between each element and an active nonmetal changes from ionic, to polar covalent, to covalent. Bonding between each element and an active metal changes from metallic to polar covalent to ionic.

14.2 TRENDS ACROSS THE PERIODIC TABLE: THE PERIOD 2 ELEMENTS

- The acid-base behavior of the common element oxide in water changes from basic to amphoteric to acidic as the bond between the element and oxygen becomes more covalent.
- Reducing strength decreases through the metals, and oxidizing strength increases through the nonmetals. In Period 2,
 - Common oxidation numbers (o.N.s) equal the A-group number for Li and Be
 - A-group number minus eight for O and F
 - Boron has several o.N.s
 - Ne has none, and
 - C and N show all possible o.N.s for their groups

Table 14.1 Trends in Atomic, Physical, and Chemical Properties of the Period 2 Elements

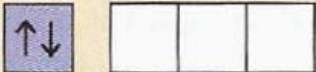
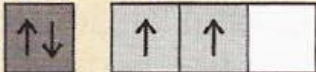
Group: Element/At. No.:	1A(1) Lithium (Li) $Z = 3$	2A(2) Beryllium (Be) $Z = 4$	3A(13) Boron (B) $Z = 5$	4A(14) Carbon (C) $Z = 6$
Atomic Properties				
Condensed electron configuration; partial orbital diagram	$[\text{He}] 2s^1$  2s 2p	$[\text{He}] 2s^2$  2s 2p	$[\text{He}] 2s^2 2p^1$  2s 2p	$[\text{He}] 2s^2 2p^2$  2s 2p
Physical Properties				
Appearance				
Metallic character	Metal	Metal	Metalloid	Nonmetal
Hardness	Soft	Hard	Very hard	Graphite: soft Diamond: extremely hard
Melting point/ boiling point	Low mp for a metal	High mp	Extremely high mp	Extremely high mp

Table 14.1 Trends in Atomic, Physical, and Chemical Properties of the Period 2 Elements

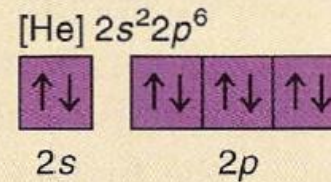
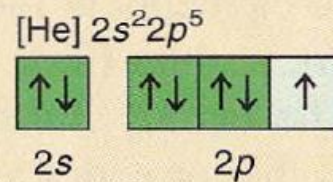
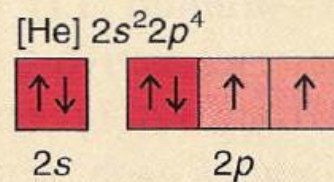
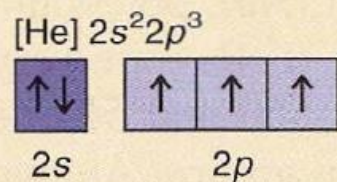
Group: Element/At. No.:	1A(1) Lithium (Li) Z = 3	2A(2) Beryllium (Be) Z = 4	3A(13) Boron (B) Z = 5	4A(14) Carbon (C) Z = 6
Chemical Properties				
General reactivity	Reactive	Low reactivity at room temperature	Low reactivity at room temperature	Low reactivity at room temperature; graphite more reactive
Bonding among atoms of element	Metallic	Metallic	Network covalent	Network covalent
Bonding with nonmetals	Ionic	Polar covalent	Polar covalent	Covalent (π bonds common)
Bonding with metals	Metallic	Metallic	Polar covalent	Polar covalent
Acid-base behavior of common oxide	Strongly basic	Amphoteric	Very weakly acidic	Very weakly acidic
Redox behavior (O.N.)	Strong reducing agent (+1)	Moderately strong reducing agent (+2)	Complex hydrides good reducing agents (+3, -3)	Every oxidation state from +4 to -4
Relevance/Uses of Element and Compounds				
	Li soaps for auto grease; thermo-nuclear bombs; high-voltage, low-weight batteries; treatment of bipolar disorders (Li_2CO_3)	Rocket nose cones; alloys for springs and gears; nuclear reactor parts; x-ray tubes	Cleaning agent (borax); eyewash, antiseptic (boric acid); armor (B_4C); borosilicate glass; plant nutrient	Graphite: lubricant, structural fiber Diamond: jewelry, cutting tools, protective films Limestone (CaCO_3) Organic compounds: drugs, fuels, textiles, etc.

5A(15)
Nitrogen (N) $Z = 7$

6A(16)
Oxygen (O) $Z = 8$

7A(17)
Fluorine (F) $Z = 9$

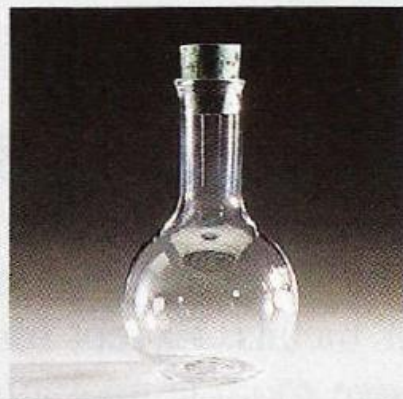
8A(18)
Neon (Ne) $Z = 10$



Nonmetal

—

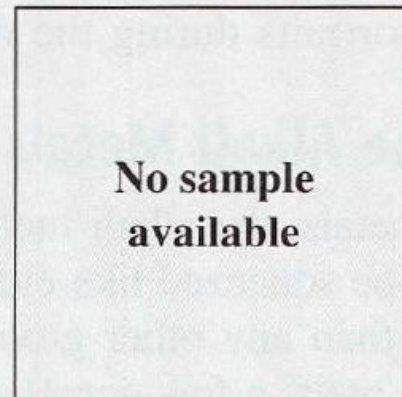
Very low mp
and bp



Nonmetal

—

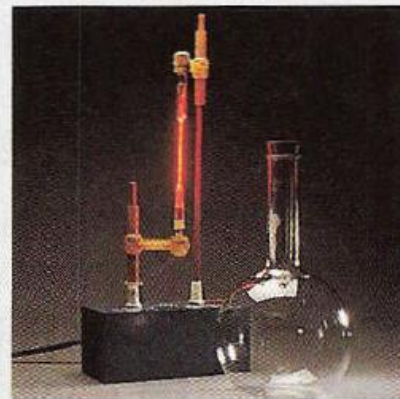
Very low mp
and bp



Nonmetal

—

Very low mp
and bp

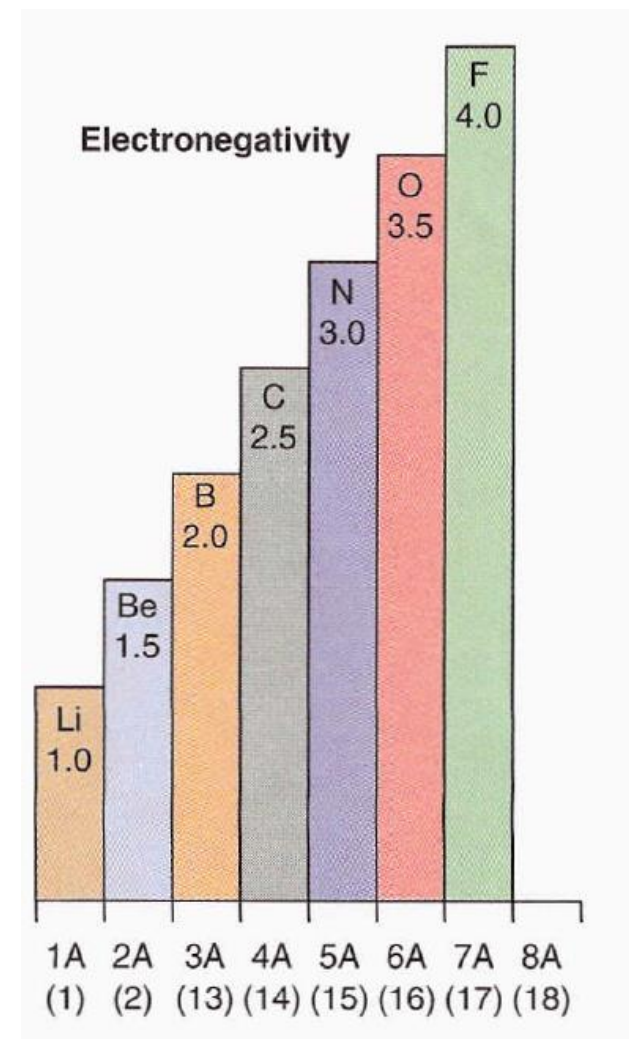
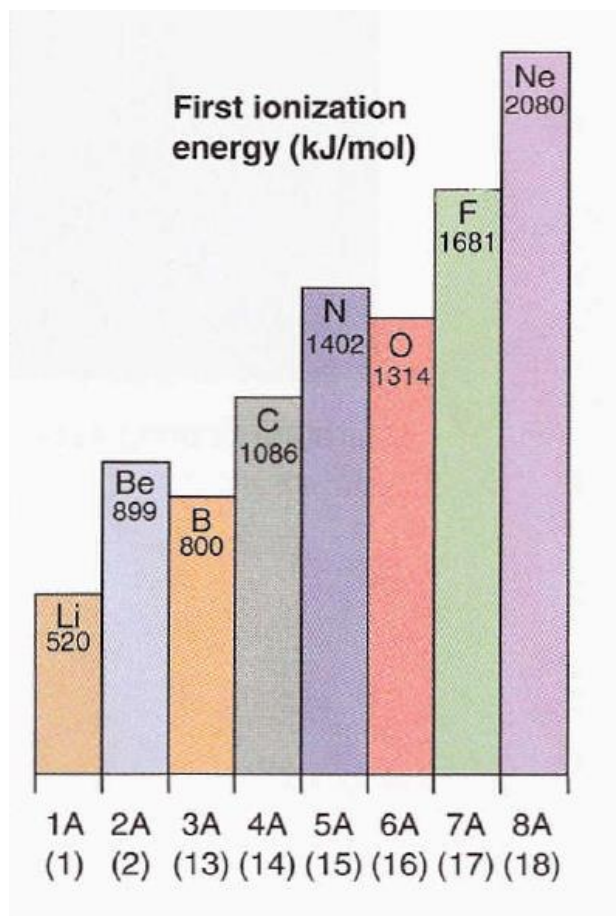
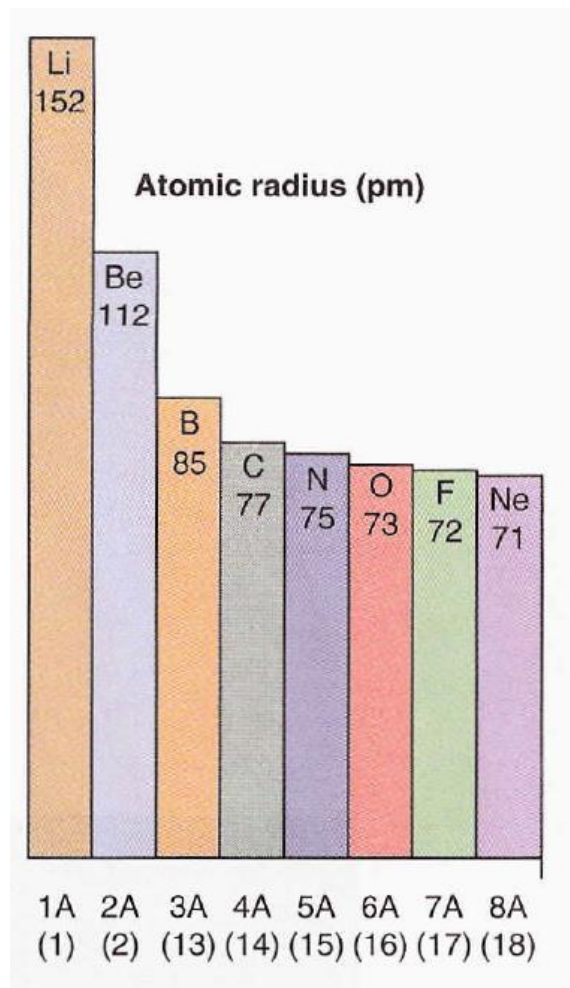


Nonmetal

—

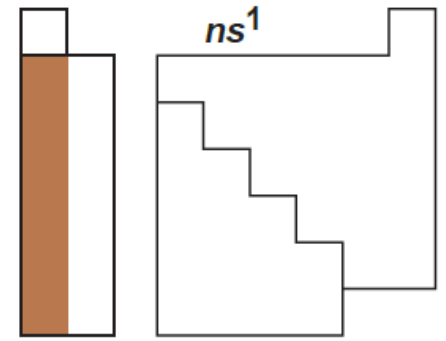
Extremely low mp
and bp

5A(15) Nitrogen (N) Z = 7	6A(16) Oxygen (O) Z = 8	7A(17) Fluorine (F) Z = 9	8A(18) Neon (Ne) Z = 10
Inactive at room temperature	Very reactive	Extremely reactive	Chemically inert
Covalent N ₂ molecules	Covalent O ₂ (or O ₃) molecules	Covalent F ₂ molecules	None; separate atoms
Covalent (π bonds common)	Covalent (π bonds common)	Covalent	None
Ionic/polar covalent; anions with active metals	Ionic	Ionic	None
Strongly acidic (NO ₂)	—	Acidic	None
Every oxidation state from +5 to -3	O ₂ (and O ₃) very strong oxidizing agents (-2)	Strongest oxidizing agent (-1)	None
Component of proteins; ammonia for fertilizers, explosives; oxides involved in manufacturing and air pollution (smog, acid rain)	Final oxidizer in residential, industrial, and biological energy production	Manufacture of coatings (Teflon); glass etching (HF); refrigerants involved in ozone depletion (CFCs); dental protection (NaF, SnF ₂)	Electrified gas in advertising signs



14.3 GROUP 1A(1): THE ALKALI METALS

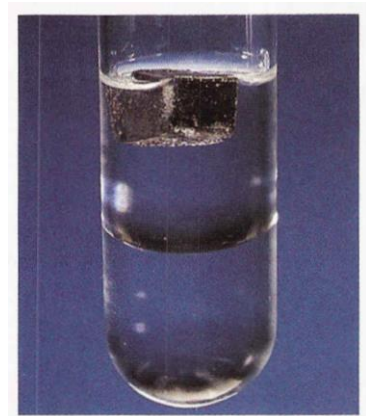
- The first group of elements in the periodic table form **alkaline (basic) solutions** when their oxides dissolved in water.
- All the elements in the group-lithium (Li), sodium (Na), potassium (K), rubidium (Rb), cesium (Cs), and rare, radioactive francium (Fr)*are **very reactive** metals.



GROUP 1A(1)

Why Are the Alkali Metals Soft, Low Melting, and Lightweight?

- Unlike most metals, the **alkali metals are soft**
 - Na has the consistency of cold butter,
 - K can be squeezed like clay.
- The **alkali metals have lower melting and boiling points** than any other group of metals.
 - Except for Li, they all melt below 100°C
 - Cs melts a few degrees above room temperature
- They **have lower densities** than most metals
 - Li floats on lightweight household oil.



Lithium floating in oil floating on water.

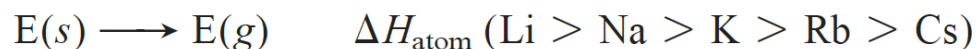
Why Are the Alkali Metals So Reactive?

- The alkali metals are extremely reactive elements.
- They are powerful reducing agents and, thus, always occur in nature as 1+ cations rather than as free metals.
 - The alkali metals reduce the halogens and form ionic solids in reactions that release large quantities of heat.
 - They reduce the hydrogen in water, reacting vigorously (Rb and Cs explosively) to form H_2 and a metal hydroxide solution.
 - They reduce O_2 in the air, and thus tarnish rapidly.

ns^1 configuration is the basis for their physical properties

1. Low heat of atomization (ΔH°_{atom}).

Consistent with the alkali metals' low melting and boiling points, their weak metallic bonding leads to low values for ΔH°_{atom} (the energy needed to convert the solid into individual gaseous atoms), which decrease down the group:



Because of this reactivity, Na and K are usually kept under mineral oil in the lab and Rb and Cs are handled with gloves under an inert argon atmosphere.

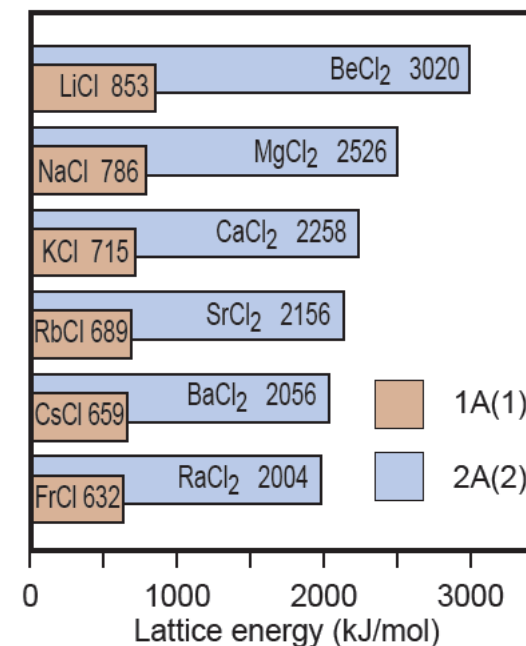
ns^1 configuration is the basis for their physical properties

2. Low IE and small ionic radius.

- Each alkali metal has the largest size and the lowest IE in its period. A great decrease in size occurs when the outer electron is lost.
 - Volume of the Li ion is less than 13% that of the Li atom!
 - Group 1A(1) ions are small spheres with considerable charge density.

3. High lattice energy.

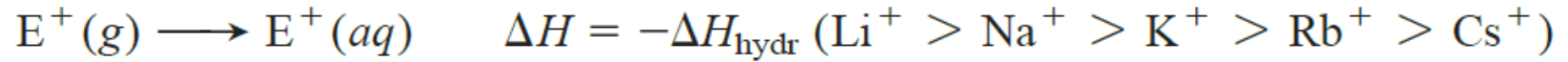
- When Group 1A(1) salts crystallize, large amounts of energy are released because the small cations lie close to the anions.
- For a given anion, the trend in lattice energy is the inverse of the trend in cation size
 - As the cation becomes larger, the lattice energy becomes smaller.



Atomic radius (pm)		Ionic radius (pm)
Li 152		Li ⁺ 76
Na 186		Na ⁺ 102
K 227		K ⁺ 138
Rb 248		Rb ⁺ 152
Cs 265		Cs ⁺ 167
Fr (~270)		Fr ⁺ 180

- Despite the strong ionic attractions in the solid, nearly all Group 1A (1) salts are water soluble.
- The attraction between the ions and water molecules creates a highly exothermic heat of hydration, and a large increase in entropy occurs when ions in the organized crystal become dispersed and hydrated in solution; together, these factors outweigh the high lattice energy.

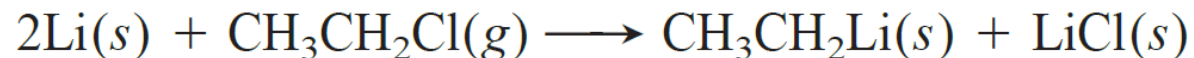
The magnitude of the hydration energy *decreases* as ionic size increases:



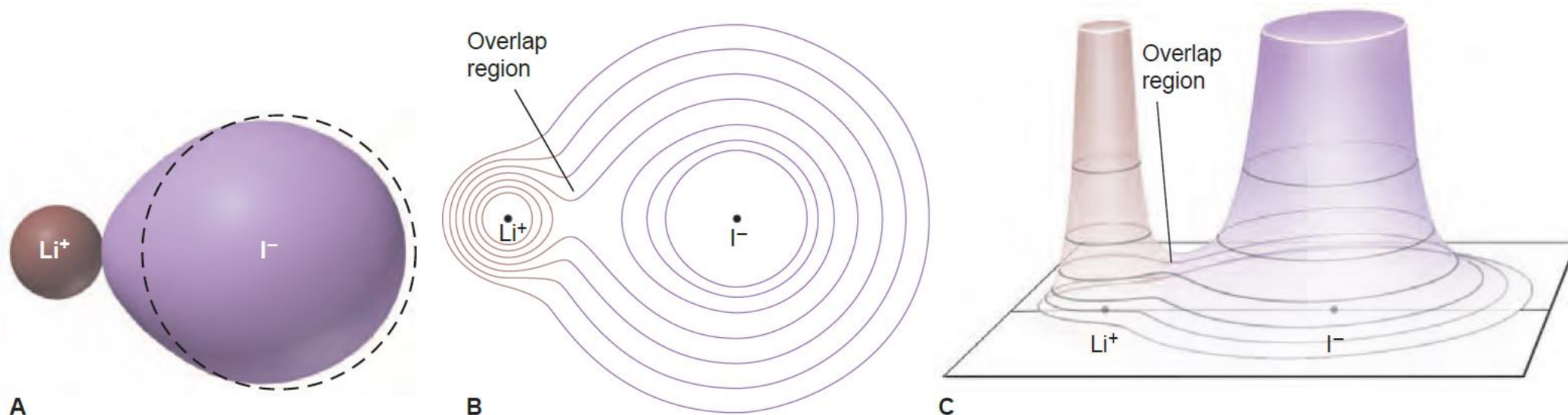
- Interestingly, the smaller ions attract water molecules strongly enough to form larger hydrated ions. This size trend has a major effect on the function of nerves, kidneys, and cell membranes because the sizes of $\text{Na}^+(aq)$ and $\text{K}^+(aq)$, the most common cations in cell fluids, influence their movement in and out of cells.

The Anomalous Behavior of Lithium

- It is the only member that forms a simple oxide and nitride, Li_2O and Li_3N , on reaction with O_2 and N_2 in air.
- Only Li forms molecular compounds with hydrocarbon groups from organic halides:



- Because of its small size, Li has a relatively high charge density.
 - Therefore, it can deform nearby electron clouds to a much greater extent than the other 1A ions can, which increases orbital overlap and gives many lithium salts significant covalent character



- Thus, LiCl, LiBr, and LiI are much more soluble in polar organic solvents, such as ethanol and acetone, than are the halides of Na and K, because the lithium halide dipole interacts with these solvents through dipole-dipole forces.
- The small, highly positive Li makes dissociation of Li salts into ions more difficult in water; thus, the fluoride, carbonate, hydroxide, and phosphate of Li are much less soluble in water than those of Na and K

Important Compounds

1. Lithium chloride and lithium bromide, LiCl and LiBr.

Because the Li ion is so small, Li salts have a high affinity for H₂O and yet a positive heat of solution. Thus, they are used in dehumidifiers and air-cooling units.

2. Lithium carbonate, Li₂CO₃.

Used to make porcelain enamels and toughened glasses and as a drug in the treatment of bipolar disorders.

3. Sodium chloride, NaCl.

Millions of tons used in the industrial production of Na, NaOH, Na₂CO₃/NaHCO₃, Na₂SO₄, and HCl or purified for use as table salt.



4. Sodium carbonate and sodium hydrogen carbonate, Na_2CO_3 and NaHCO_3 .

Carbonate used as an industrial base and to make glass. Hydrogen carbonate, which releases CO_2 at low temperatures (50°C to 100°C), used in baking powder and in fire extinguishers.



5. Sodium hydroxide, NaOH .

Most important industrial base; used to make bleach, sodium phosphates, and alcohols.

6. Potassium nitrate, KNO_3 .

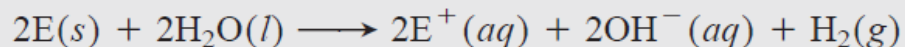
Powerful oxidizing agent used in gunpowder and fireworks



Important Reactions

The reducing power of the alkali metals (E) is shown in reactions 1 to 4. Some industrial applications of Group 1A(1) compounds are shown in reactions 5 to 7.

1. The alkali metals reduce H in H_2O from the +1 to the 0 oxidation state:

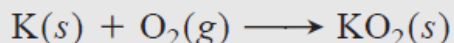
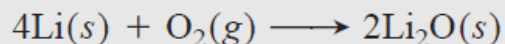


The reaction becomes more vigorous down the group (*see photo*).



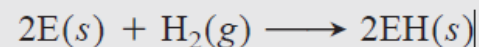
Potassium reacting
with water

2. The alkali metals reduce oxygen, but the product depends on the metal. Li forms the oxide, Li_2O ; Na forms the peroxide, Na_2O_2 ; K, Rb, and Cs form the superoxide, EO_2 :



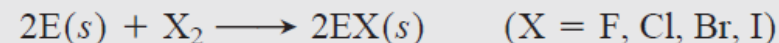
In emergency breathing units, KO_2 reacts with H_2O and CO_2 in exhaled air to release O_2 .

3. The alkali metals reduce hydrogen to form ionic (saltlike) hydrides:



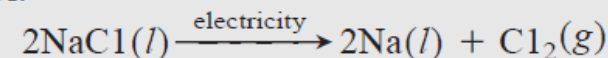
NaH is an industrial base and reducing agent that is used to prepare other reducing agents, such as NaBH_4 .

4. The alkali metals reduce halogens to form ionic halides:

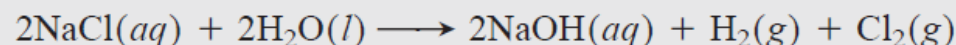


5. Sodium chloride is the most important alkali metal halide.

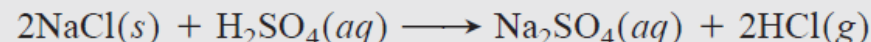
(a) In the Downs process for the production of sodium metal (*Section 22.4*), reaction 4 is reversed by supplying electricity to molten NaCl :



(b) In the chlor-alkali process (*Section 22.5*), $\text{NaCl}(aq)$ is electrolyzed to form several key industrial chemicals:

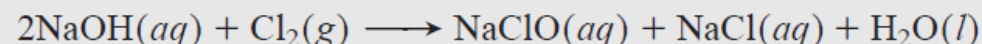


(c) In its reaction with sulfuric acid, NaCl forms two major products:



Sodium sulfate is important in the paper industry; HCl is essential in steel, plastics, textile, and food production.

6. Sodium hydroxide is used in the formation of bleaching solutions:



7. In an ion-exchange process (*Chemical Connections*, p. 542), water is “softened” by removal of dissolved hard-water cations, which displace Na^+ from a resin:

